Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/30/2025
PI Name:	Doherty, Colleen Ph.D.		
Project Title:	How Do Carbon Fixing Strategies Affect Nutritional Content Under High CO2? A Comparison of C3 vs. C4 Microgreens		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27695-7622	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies
Start Date:	12/08/2022	**End Date:**	12/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2024 per NSSC information (Ed., 12/17/24). NOTE: End date changed to 12/06/2024 per NSSC information (Ed., 10/9/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Perera, Imara Ph.D. (North Carolina State University)		
Grant/Contract No.:	80NSSC23K0344		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Several carbon capture mechanisms have emerged in plant systems that provide unique advantages to plants depending on their environment. For example, while most plants use C3 photosynthesis, C4 and Crassulacean acid metabolism (CAM) carbon capture mechanisms can increase water use efficiency or temperature tolerance. These advantages have been well-characterized in the atmospheric carbon dioxide (CO2) levels on Earth, but in enclosed human habitats such as those needed for long-term spaceflight, CO2 levels far exceed that of the Earth's atmosphere. Altered CO2 levels are known to affect nutritional content and water use efficiency; however, much of the prior research has used CO2 levels below that on enclosed human habitats. This proposed work would examine how high CO2 levels affect plant physiology and nutritional content of edible microgreens that use different photosynthetic mechanisms: C3, C4, and CAM. We will monitor physiological characteristics and the nutritional profile across different CO2 levels for select microgreen species (with C3, C4, and CAM photosynthesis). The combined effects of altered CO2 levels and other spaceflight relevant stresses such as water availability will be examined to understand if these different photosynthetic mechanisms can provide advantages to enhance plant productivity in space environments. These results would provide important baseline information on plant nutrition and performance that is needed for planning long-term space missions and thus would address the following objectives of the solicitation and NASA program goals: Decadal Survey-Priority 3: A systematic suite of plant biology experiments to elucidate mechanisms by which plants respond and adapt to spaceflight, and to facilitate their eventual use in Bioregenerative Life Support Systems; PB-1 How does gravity affect plant growth, development & metabolism (e.g. photosynthesis, reproduction, lignin formation, plant defense mechanisms); and PB-3 How can horticultural approaches for sustained production of edible crops in space be both improved and implemented (especially as related to water and nutrient provision in the root zone)?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	CO2 levels impact plant physiology. However, extreme CO2 levels, as are found in contained human environments such as on the ISS, are not commonly found in plant-growing areas on Earth. Understanding how plants perform in these environments and the effects on physiology and nutrition could affect decisions about using such environments to increase crop production on Earth.
Task Progress:	The new design of the extreme CO2 (eCO2) chamber was completed, and the new chamber has tight control over the CO2 levels in our control condition. Issues with chamber humidity variation were resolved. The eCO2 chamber design addresses an issue where, at night, the plants' increase in CO2 emissions substantially altered the CO2 levels. The eCO2 chamber design allows venting of chamber air to the outside to lower CO2 levels to maintain CO2 levels within a set threshold during both the day and nighttime hours. Other parameters including temperature, humidity, and air pressure were consistent between the eCO2 chamber and the control chamber. In conjunction with the metabolite analysis, we investigated the effects of International Space Station (ISS)-level CO_2 conditions on plant physiology – specifically, growth, biomass, and photosynthetic parameters. Issues with chamber humidity, pathogens, and germination issues that were specific to growth in these chambers were resolved and enabled the collection of sufficient samples to evaluate physiological responses. We collected detailed physiological data from 16 plants per box – eight from the outer edge and eight from the inner portion. Measurements included shoot mass, hypocotyl length and width, number of leaves, and the leaf area of individual leaves longer than 5 mm. This sampling strategy yielded data from 32 plants per CO_2 condition in each experimental run. After excluding boxes with signs of disease, a total of 192 radish plants were analyzed under indoor ambient CO_2 and 176 under elevated CO_2 (eCO_2). For amaranth, 128 plants were analyzed under ambient CO_2 and 144 under eCO_2. Water usage, pH, and the combined weight of root and substrate were measured at the whole-box level. Following the exclusion of contaminated or spilled boxes, 11 boxes were sampled for each CO_2 condition for radish, while 8 ambient and 9 eCO_2 boxes were analyzed for amaranth. No significant differences were observed between CO_2 treatments for shoot mass, leaf area, leaf number, pH, or water use in either species. However, elevated CO_2 had subtle but statistically significant effects on shoot morphology. Both species exhibited changes in hypocotyl dimensions: radish developed thicker hypocotyls, while amaranth produced longer ones under eCO_2. Additionally, radish plants showed a significant increase in root mass under eCO_2, an effect not observed in amaranth. To evaluate the physiological effects of eCO_2, we measured carbon assimilation (A) and stomatal conductance (gsw) on the five largest leaves per box, resulting in ten leaves analyzed per CO_2 treatment per replicate. Measurements for eCO_2 plants that were grown in 3,000 ppm CO_2 were conducted at 2000 ppm – the maximum CO_2 concentration our LI-6800 system could reliably maintain. Indoor ambient controls were measured at 450 ppm CO_2, consistent with daytime levels recorded in the ambient chambers. In radish, plants grown under elevated CO_2 exhibited significantly higher carbon assimilation rates compared to ambient controls. Mean assimilation for ambient-grown radish was 19.97 µmol CO_2 m^{-2} s^{-1}, while eCO_2-grown plants averaged 22.54 µmol CO_2 m^{-2} s^{-1} . Amaranth showed a similar trend: ambient-grown plants had a mean assimilation rate of 13.60 µmol CO_2 m^{-2} s^{-1}, which increased significantly to 15.48 µmol CO_2 m^{-2} s^{-1} under eCO_2. The overall assimilation rates were lower in amaranth than in radish. Our findings challenge some common assumptions about plant responses to elevated CO_2 (eCO_2). While radish is known to require more water than amaranth, it still exhibited higher stomatal conductance at harvest. Surprisingly, radish proved more water efficient when measured as milliliters of water used per gram of shoot biomass, likely due to amaranth's slower growth and prolonged exposure to substrate evaporation. Contrary to prior reports that elevated CO_2 stimulates increased leaf area (Huang and Xu, 2015), neither species in our study showed significant changes in shoot mass or leaf area at ISS-level eCO_2. In fact, radish showed a slight, non-significant reduction in mean leaf area under these conditions. This contrast with previous studies may stem from the lower eCO_2 concentration used in most studies. The age of the plants evaluated may also play a role. Our focus was on early microgreens, but most studies are on older, established plants. For example, Usuda and Shimogawara (1998) observed increased radish leaf area at 750 ppm eCO_2 – substantially lower than the 3,000 ppm used in our experiments.

Bibliography Type:	Description: (Last Updated: 05/25/2026)
Abstracts for Journals and Proceedings	Toennisson A, Horton TB, Perera IY, Doherty CJ. "Comparing the effects of elevated carbon dioxide on radish and amaranth microgreens." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Toennisson A, Horton TB, Perera IY, Doherty CJ. "Comparing the effects of extreme eCO2 on radish and amaranth microgreens." 2025 Annual Meeting of the Southern Section of the American Society of Plant Biologists (SS-ASPB), Raleigh, NC, March 28-30, 2025. Poster. Abstracts. 2025 Annual Meeting of the Southern Section of the American Society of Plant Biologists (SS-ASPB), Raleigh, NC, March 28-30, 2025. , Mar-2025
Articles in Peer-reviewed Journals	Toennisson TA, Horton TB, Rueckeis S, Chiera J, Cho S, Perera IY, Doherty CJ. "Building and operating a low-cost elevated carbon dioxide growth chamber to evaluate microgreen physiology under spaceflight-relevant CO2 levels." Journal of Visualized Experiments (JoVE). 2026 Apr 17(230):e70050. https://doi.org/10.3791/70050 , Apr-2026
Significant Media Coverage	Toennisson TA, Horton TB, Rueckeis S, Chiera J, Cho S, Perera IY, Doherty CJ. "Building and operating a low-cost elevated carbon dioxide growth chamber to evaluate microgreen physiology under spaceflight-relevant CO2 levels. Video demonstration." Journal of Visualized Experiments (JoVE). 2026 Apr 17(230):e70050. https://doi.org/10.3791/70050-v , Apr-2026